Math 150A Section 8 and 16
Spring 2020
Practice Exam I
February 13, 2020
Time Limit: 1 hour 50 Minutes

Name (Print): ____________________

Student ID: ____________________

This exam contains 10 pages (including this cover page) and 9 problems. Check to see if any pages are missing. Enter all requested information on the top of this page, and put your initials on the top of every page, in case the pages become separated.

You may *not* use your books or notes on this exam. However, you may use a *basic* calculator.

You are required to show your work on each problem on this exam. The following rules apply:

- **Organize your work**, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little credit.
- **Mysterious or unsupported answers will not receive full credit**. A correct answer, unsupported by calculations, explanation, or algebraic work will receive no credit; an incorrect answer supported by substantially correct calculations and explanations might still receive partial credit. This especially applies to limit calculations.
- If you need more space, use the back of the pages; clearly indicate when you have done this.
- **Box Your Answer** where appropriate, in order to clearly indicate what you consider the answer to the question to be.

Do not write in the table to the right.

Problem	Points	Score
1	10	
2	12	
3	10	
4	6	
5	8	
6	10	
7	10	
8	7	
9	7	
Total:	80	

1. (10 points) **TRUE or FALSE!** Write

   (a) if $f(1) = 0$ and $g(1) = 3$, then $g(f(1)) = 3$

   (b) if $f(3)$ does not exist, then $\lim_{x\to 3} f(x)$ does not exist

   (c) if $f(x)$ is differentiable at 2, then $f(x)$ is continuous at 2

   (d) the functions $f(x) = \frac{x+1}{x+1}$ and $g(x) = 1$ are the same

   (e) $(f(x)g(x))' = f'(x)g'(x)$

2. (12 points) For each of the following limits, calculate the limit or explain why it does not exist.

(a)

$$\lim_{x \to 2} \frac{2x}{x+2}$$

(b)

$$\lim_{x \to 3} \frac{\sqrt{x+1}-2}{x^2-9}$$

(c)

$$\lim_{x \to 0} \frac{\frac{1}{3+x} - \frac{1}{3-x}}{x}$$

(d)

$$\lim_{x \to \infty} \sqrt{x-2} - \sqrt{x}$$

3. (10 points) Calculate the limit of each of the following functions, or explain why the limit does not exist.

$$f(x) = \frac{2x^2 + x - 3}{x^2 - 1}$$

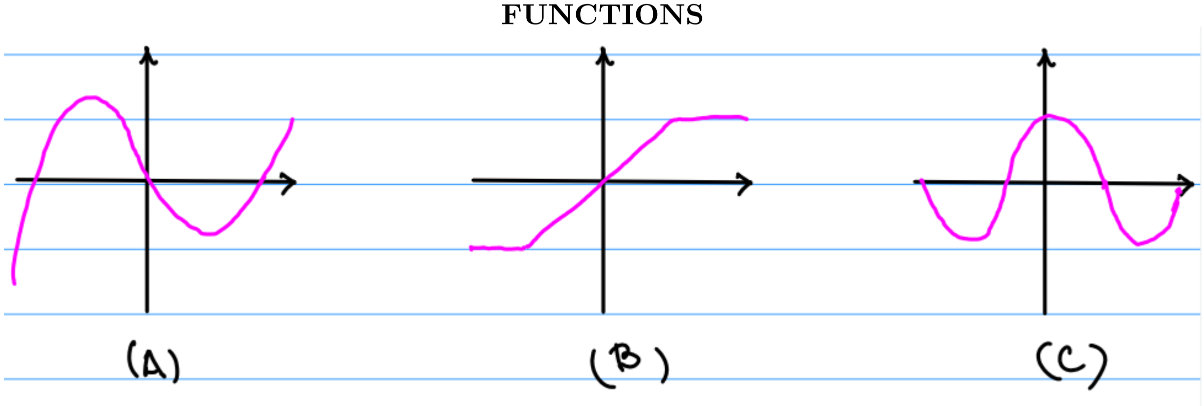
(a) $\lim_{x \to \infty} f(x)$

(b) $\lim_{x \to -1} f(x)$

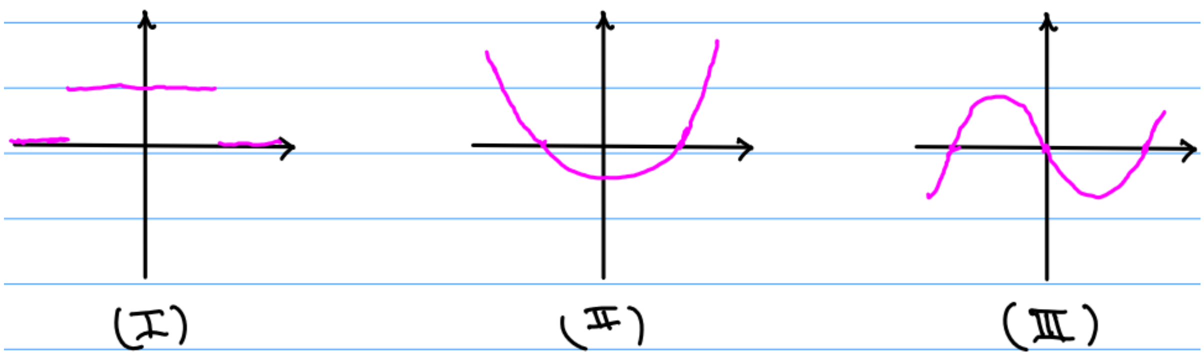
(c) $\lim_{x \to 1} f(x)$

4. (6 points) Match each graph up with the corresponding graph of its derivative.

FUNCTIONS



DERIVATIVES



5. (8 points) Consider functions $f(x)$, $g(x)$, and $h(x)$ whose values satisfy

x	1	2	3	4	5	6	7
$f(x)$	1	1	2	3	5	8	13
$f'(x)$	0	1	3	2	7	9	4

x	1	2	3	4	5	6	7
$g(x)$	8	6	7	5	3	0	9
$g'(x)$	0	1	0	1	2	3	5

x	1	2	3	4	5	6	7
$h(x)$	2	3	5	7	1	13	17
$h'(x)$	4	3	2	1	2	3	4

Evaluate the following expressions

(a) $F(5)$ for $F(x) = 2f(x) + 3g(x) + h(x)$

(b) $F'(5)$ for $F(x) = 2f(x) + 3g(x) + h(x)$

(c) $F(2)$ for $F(x) = h(g(f(4)))$

(d) $F'(4)$ for $F(x) = x^3 + 3x + h(x)$

6. (10 points) Calculate the derivative of each of the following functions. Carefully show your work, explaining which differentiation rules you are using along the way.

   (a) $f(x) = x + \cos(x)$

   (b) $g(t) = \frac{\sqrt{t}+1}{t^2}$

   (c) $h(\theta) = (\theta + \theta^3)^2$

7. (10 points) (a) Write down the ϵ,δ-definition of a limit.

(b) Let $f(x) = \sqrt{x}$. Find a value of

$$\delta > 0$$

such that if

$$|x - 4| < \delta \quad \text{implies} \quad |f(x) - 2| < 0.1$$

8. (7 points) Using only the formal definition of a derivative and any limit laws, calculate the derivative of

$$f(x) = \sqrt{2x + 1}.$$

9. (7 points) Let a and b be constants and consider the function

$$f(x) = \begin{cases} x^2 + x & 0 \le x < 1 \\ a + bx & x \ge 1 \end{cases}$$

(a) For which values of a and b is $f(x)$ continuous at $x = 1$?

(b) For which values of a and b is $f(x)$ differentiable at $x = 1$?